

Decision Tree — 5 Numerical Questions

Instructions: Use the ID3 algorithm. Show all calculations clearly using Entropy and Information Gain. Use $\log_2$ wherever required.

Question 1 — Entropy & Information Gain

A dataset contains 14 customers. The target variable indicates whether a customer will purchase a product.

Age Group	Income	Purchase
Young	High	No
Young	Low	Yes
Young	Medium	Yes
Young	High	No
Middle	High	Yes
Middle	Low	Yes
Middle	Medium	Yes
Middle	High	Yes
Senior	High	No
Senior	Low	No
Senior	Medium	Yes
Senior	High	No
Middle	Low	Yes
Young	Medium	No

Using the ID3 algorithm:

- Calculate the entropy of the target variable.
- Calculate Information Gain for Age Group.
- Calculate Information Gain for Income.
- Select the attribute that should be the root node.

Question 2 — Selecting the Best Root Node

A company wants to predict whether an employee will **Leave the Company** based on Job Satisfaction, Salary Level, and Overtime.

Job Satisfaction	Salary	Overtime	Leave
Low	Low	Yes	Yes
Low	Medium	Yes	Yes
Medium	Low	No	No
High	High	No	No
High	Medium	No	No
Medium	Medium	Yes	Yes
Low	High	Yes	Yes
High	Low	No	No
Medium	High	No	No
Low	Medium	No	No

- Calculate the entropy of the dataset.
- Calculate Information Gain for Job Satisfaction, Salary, and Overtime.
- Identify the best attribute for the root node.
- Draw the resulting first-level decision tree.

Question 3 — Build a Decision Tree

Consider the following dataset for predicting whether a person will **Play Tennis**.

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rain	Mild	High	False	Yes
Rain	Cool	Normal	False	Yes
Rain	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rain	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes

Using the ID3 algorithm:

- Calculate the entropy of the target variable.
- Calculate Information Gain for Outlook, Temperature, Humidity, and Windy.
- Select the root node.
- Construct the decision tree up to the next level wherever required.

Question 4 — Decision Tree with a Continuous Attribute

A bank wants to predict whether a loan will be **Approved** based on Age, Income, and Credit Score. The following training data is available:

Age (years)	Income (■000)	Credit Score	Loan Approved
25	30	Low	No
28	45	Low	No
35	50	Medium	Yes
40	60	Medium	Yes
45	35	High	Yes
50	40	High	Yes
52	25	Low	No
55	30	Medium	No
60	70	High	Yes
23	55	Medium	Yes
31	28	Low	No
48	65	High	Yes

Using the ID3 Decision Tree algorithm:

a) Calculate the entropy of the target variable **Loan Approved**.

b) For the continuous attribute **Age**, evaluate the following candidate split points:

$t \in \{27, 30, 33, 37, 42, 47, 51, 53, 57, 58\}$

For each t , consider the binary split: **Age** $\leq t$ and **Age** $> t$. Calculate the weighted entropy and Information Gain for each split.

c) Determine the best split point for Age.

d) Calculate the Information Gain for the categorical attribute **Credit Score**.

e) Compare the best Age split with Credit Score and determine the root node/split.

f) Draw the resulting first-level Decision Tree.

Question 5 — Entropy, Information Gain & Final Prediction

A bank has collected the following data to predict whether a customer will **Default on a Loan**.

Credit History	Income	Existing Loan	Default
Good	High	No	No
Good	Low	Yes	Yes
Good	Medium	No	No
Poor	Low	Yes	Yes
Poor	High	Yes	Yes
Poor	Medium	No	Yes
Good	High	Yes	No
Poor	High	No	No
Good	Medium	Yes	Yes
Poor	Low	No	Yes
Good	Low	No	No
Poor	Medium	Yes	Yes

Using the ID3 Decision Tree algorithm:

- Calculate the entropy of the target variable.
- Calculate Information Gain for Credit History, Income, and Existing Loan.
- Select the best root node.
- Construct the first two levels of the decision tree.
- Predict whether a customer with Poor Credit History, High Income, and No Existing Loan will default.